

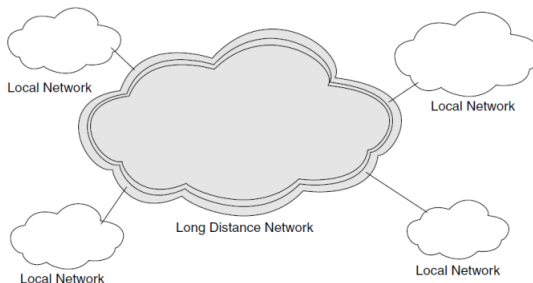
CHAPTER THREE

Wireless Telecommunication Technologies

- Fundamental of telecommunication Technologies
 - Introductory issues in telecom technology, Network Topology, QoS, Models of Telecom channels.
- Cellular Telecom Technology
 - First Generation CT (1G), Second Generation CT (2G), Third Generation CT (3G), Fourth Generation CT (4G), Fifth Generation CT (5G)

Introductory issues in telecom technology

- An overall telecom network consists of local networks interconnected by a long-distance network.
- It is usually regulated by a **government authority** or may be a **government monopoly**, although there is a notable trend toward privatization.

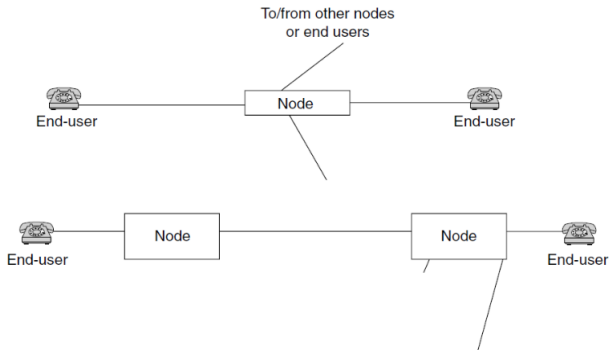


End users, Nodes, and Connectivity

- **End-users** - provide the inputs to the network and are recipients of network outputs.
 - Employs what is called an I/O, standing for input/output (device)
 - An I/O may be a PC, computer, telephone instrument, cellular/ PCS telephone or combined device, facsimile, or conference TV equipment.
 - Usually connect to **node**.
 - We will call a **node**, a point or junction in a transmission system where lines and trunks meet.
- A node usually carries out a switching function.
- In this case a network interface unit is used, through which one or more end-users may be connected.
- A **connectivity** connects an end-user to a node.

End-users, Nodes, and Connectivity

- Illustrating the functions of end-users, nodes, and connectivity.



How does telephone call happens ?

- There are three sequential stages to a telephone call:
 - 1 Call setup
 - 2 Information exchange and
 - 3 Call take down.
- Call **setup** is the stage where a circuit is established and activated - *facilitated by signaling*.
- It is initiated by the **calling subscriber (user)** going off-hook.
- The calling subscriber (user) now knows that she/ he may start dialing digits or pushing buttons on the *subscriber instrument*.
- Each button is associated with a digit. There are 10 digits, 0 through 9.
- A telephone end instrument connected through a subscriber loop to a local serving exchange.

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- What if the called subscriber and the calling subscriber are in the same local area:
 - Only nine digits (our case?) need be dialed.
 - These nine digits represent the telephone number of the called subscriber (user).
 - This type of signaling, the dialing of the digits, is called address signaling
 - The digits actuate control circuits in the local switch, allowing a connectivity to be set up.
 - No further action need be taken - same local area.
- A connection is made to the called subscriber line and the switch sends a special ringing signal.
- This audible ringing is called alerting, another form of signaling.
- This all is information exchange (phase 2), Phase 3 - terminates the call! Both subscriber loops are now idle.

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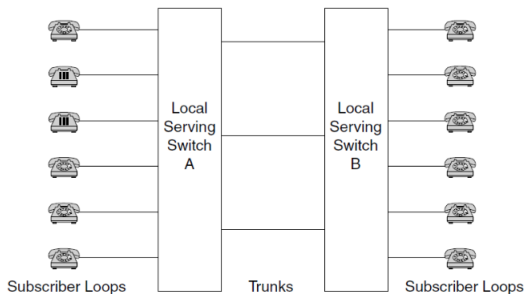
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- What If a third user tries to call either subscriber during stages 2 and 3:
 - She/he is returned a busy-back by the exchange (serving switch).
 - This is the familiar "busy signal", a tone with a particular cadence.
 - The return of the busy-back is a form of signaling called call-progress signaling.
- If a subscriber wishes to call another telephone subscriber outside the local serving area of her/ his switch:
 - The call setup will be similar as before.
 - Except that at the calling subscriber serving switch the call will be connected to an outgoing trunk.
- IEEE defines a trunk as "a transmission path between exchanges or central offices."
- Subscriber loops connect telephone subscribers to their local serving exchange; trunks interconnect exchanges (switches).

End-users, Nodes, and Connectivity...

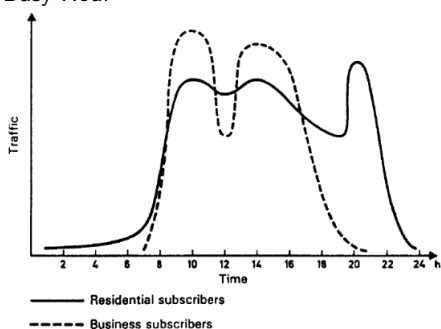
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- The telecom network is very inefficient, especially in developing countries.
- This inefficiency stems from the number of circuits and the revenue received per circuit.
- The telecom network would approach 100% the circuits were used all the time.
- **Fact** - the telecom network approaches total capacity utilization for only several hours during the working day.
 - 1 From 10P.M.-7A.M. capacity utilization may be 2
 - 2 The network is dimensioned (sized) to meet the period of maximum usage demand.
 - 3 This period is called the busy hour (BH).
 - 4 Periods - traffic demand is maximum, morning and afternoon.

Busy Hour and Grade of Service ...

- Busy Hour



- Networks are sized/ dimensioned for a traffic load expected during the busy hour.
- The sizing is based on probability, usually expressed as a decimal or percentage.

- That probability percentage or decimal is called **grade of service**.
 - ① IEEE defines grade of service as "the proportion of total calls, usually during the busy hour, that cannot be completed immediately or served within a prescribed time."
 - ② Grade of service and blocking probability are synonymous.
 - ③ Blocking probability objectives are usually stated as $B = 0.01$ or 1
 - ④ This means that during the busy hour 1 in 100 calls can be expected to meet blockage.

Simplex, Half Duplex, and Full Duplex

- These are operational terms, and they will be used throughout the telecom course.
 - 1 **Simplex**: is one-way operation; there is no reply channel provided.
 - Ex: Radio and television broadcasting.
 - 2 **Half-duplex**: is a two-way service. It is defined as transmission over a circuit capable of transmitting in either direction, but only in one direction at a time.
 - Ex: Walkie-Talkie
 - 3 **Full duplex or just duplex**: defines simultaneous two-way independent transmission on a circuit in both directions.
 - Ex: All telecom-type circuits

- QoS appears at the outset to be an intangible concept.
- However, it is very tangible for a telephone subscriber unhappy with his or her service.
- The concept of service quality must be covered early in an all-encompassing text on telecom.
- System designers should never once lose sight of the concept - at any level of responsibility!
- QoS means how happy the telephone company (or other common carrier) is keeping the customer. For instance:
 - About half the time a customer dials, the call goes awry.
 - The caller cannot get a dial tone or cannot hear what is being said by the party at the other end.
- All these have an impact on quality of service.
- QoS is an important factor in many areas of the telecom business and means different things to different people

Is there any measuring parameter then ?

- In the old days of telegraphy - the number of service messages received at the switching center.
- In modern telephony - **Service Observation!**
- The transmission engineer calls QoS **customer satisfaction**, which is commonly measured by how well the customer can hear the calling party.
- The unit for measuring how well we can hear a distant party on the telephone is loudness rating, measured in decibels (dB).
- From the network and switching viewpoints:
- The percentage of lost calls (due to blockage or congestion) during the BH certainly constitutes another measure of service quality.
- Denominated grade of service. One target figure for grade of service is 1 in 100 calls lost during the busy hour.

- Other elements to be listed under QoS are:
 - ① Delay before receiving dial tone (dial tone delay);
 - ② Post dial(ing) delay - primary measure of signaling quality;
 - ③ Availability of service tones [e.g., busy tone, telephone out of order, time out, and all trunks busy (ATB)];
 - ④ Correctness of billing;
 - ⑤ Reasonable cost of service to the customer;
 - ⑥ Responsiveness to servicing requests;
 - ⑦ Responsiveness and courtesy of operators; and
 - ⑧ Time to installation of a new telephone and, by some, the additional services offered by the telephone company.
- Depending on the service quality goal, each item will have an impact on the design of a telecommunication system.

- Standardization is vital in telecommunications.
- A rough analogy it allows worldwide communication because we all "speak a standard language."
- There are int., regional, and national standardization agencies.
- Two international agencies that impact telecom:
 - International Telecommunication Union (ITU) - produced more than 1000 standards.
 - International Standardization Organization (ISO)- issued a number of important data communication standards.
- We note one important regional organization:
 - European Telecommunication Standardization Institute (ETSI)- responsible for eg. a principal cellular radio specification.
- There are numerous national standardization organizations.
Ex: IEEE

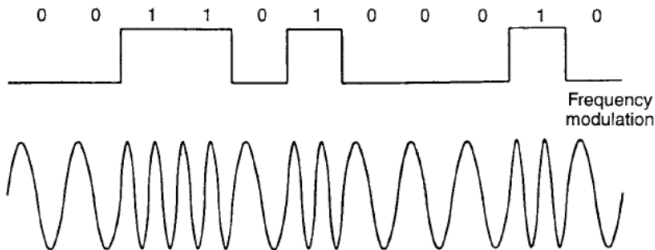
Modes of Telecom Channels

- At the transmitting side of a telecommunication link a radio carrier is generated.
- The carrier is characterized by a frequency.
- This single radio frequency carries no useful information for the user. Useful information may include voice, data, or image.
- **Modulation** is the process of impinging that useful information on the carrier.
- **Demodulation** is the recovery of that information from the carrier at the distant end near the destination user.
- There are three generic forms of modulation:
 - 1 Amplitude Modulation (AM),
 - 2 Frequency Modulation (FM), and
 - 3 Phase Modulation (PM).

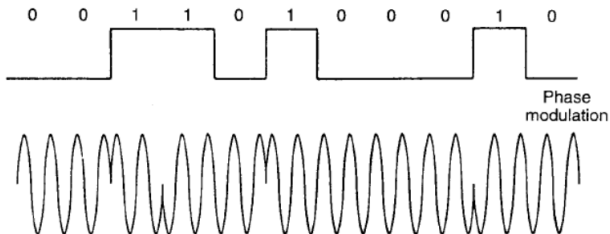
Modes of Telecom Channels...

- **Amplitude modulation** is where a carrier is varied in amplitude in accordance with the information baseband signal.
- **Frequency modulation** is where a carrier is varied in frequency in accordance with the baseband signal
- **Phase modulation** is where a carrier is varied in its phase in accordance with the information baseband signal.
- Prior to 1960, all transmission systems were analog.
- Today, all telecommunication systems are digital, except for the preponderance of subscriber access lines.

- Illustration of frequency modulation, where the modulating signal is the binary digital sequence 00110100010, an electrical baseband signal..



- Illustration of phase modulation, where the modulating signal is the binary digital sequence 00110100010, an electrical baseband signal.



0-Generation Telecom tech

④ Mobile Telephony System

- The first public mobile telephone system → Mobile Telephone System (MTS)
- It was also known as Car-based mobile Telephony
 - **Mobile transceivers** of MTS were very big and could be carried only by vehicles.
- MTS was Analog system
 - meaning that it processed voice information as a continuous waveform.
- The system was half-duplex
 - Communication parties @ a time could either speak or listen.
 - To switch between the two modes, users had to push a specific button on the terminal.
- MTS utilized a Base Station (BS) with a single high-power transmitter.
- Base Stations utilized the same frequencies.
- Due to **power limitations**, mobile units transmitted not directly to the BS but to receiving sites

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Mobile Telephon System (MTS)...

- How a call happen in MTS ?
 - The caller first called a special number to connect to an MTS operator.
 - The caller informed the operator of the mobile subscriber's number.
 - Then the operator searched for an idle channel in order to relay the call to the mobile terminal.
- Major limitations:
 - Manually Switching of calls.
 - limited channels are available (3 channels)
- Improved Mobile Telephone System (IMTS)
 - Utilized *automatic call switching*
 - Enhanced its mode of transmission → *Full-duplex*
 - The *intermediation of the operator* in a call Eliminated.
 - Number of channel increased to 23
 - IMTS has **problem**:
 - providing a small capacity Because spectrum usage.
 - Interference to adjacent systems due to *large power of BS transmitter*.

Generation of Wireless/Telecom technologies

1G: Where wireless communication began

- 1G refers to the first-generation of wireless telephone technology or mobile telecommunications.
- These are the analog telecommunications standards that were introduced in the 1980s
- and continued until being replaced by 2G digital telecommunications.
- Analog signals were used to transmit data.
- The first generation of cellular systems (1G systems) - concept designed
- **Refer the 1G standards in different countries?**
- 1G has characterized by :
 - Low Speed (2.4kbps)
 - Quality of voice $\Rightarrow$ **POOR**
 - Large phones with limited battery life
 - Security

- Limitation of 1G...
 - Limited coverage;
 - Full analog mode of communication, hence inefficient use of the spectrum;
 - Low capacity, FDMA technique does not maximize system capacity;
 - Different 1G systems are incompatible with one another, due to different frequency ranges of the systems;
 - No roaming supported between different operators;
 - Weak security on air interface, no support for encryption;
 - No Mobile Assisted Handover and hence more burden on the Mobile Switching Center (MSC).

AMPS (Advanced Mobile Phone System)

- Invented by Bell Labs and first installed in the United States in 1982.
- In AMPS, the cells are typically 10 to 20 km across;
- Each cell uses some set of frequencies not used by any of its neighbors.
- an AMPS system might have 100 10-km cells in the same area and be able to have 10 to 15 calls on each frequency, in widely separated cells.

- Every cell has BS. where connected to Mobile Switching Center(MSC).
- AMPS uses an arrangement is known as FDD (Frequency Division Duplex).
 - The system uses 832 full-duplex channels, each consisting of a pair of simplex channels.
 - The *832 simplex* channels from 824 to 849 MHz are used for *mobile to base station transmission*
 - 832 simplex channels from 869 to 894 MHz are used for *base station to mobile transmission*.
 - Each of these simplex channels is 30 kHz wide.
- The 832 different Channels categorized in to 4:
 - 1 Control Channel $\Rightarrow$ Manages the system from base to mobile.
 - 2 Paging Channels $\Rightarrow$ alert mobile users to calls for them.
 - 3 Access Channels $\Rightarrow$ are used for call setup and channel assignment.
 - 4 Data Channels $\Rightarrow$ carry voice, fax, or data.

Questions?!